1. Solve the following equation algebraically and check graphically.

   $8966 = e^{2x}$

   x ≈ [ ]
   (Do not round until the final answer. Then round to three decimal places as needed.)

2. Solve the logarithmic equation.

   $\log_2 x = 2$

   x = [ ]
   (Simplify your answer. Type an exact answer, using e as needed.)

3. Solve the logarithmic equation.

   $4 + 5 \ln x = 44$

   x = [ ]
   (Simplify your answer. Type an exact answer, using e as needed.)

4. Solve the equation algebraically and check graphically.

   $2e^{8x} = 1246$

   x ≈ [ ]
   (Round to the nearest thousandth as needed.)

5. Solve the equation algebraically and check graphically.

   $3000 = 200\left(10^x\right)$

   x ≈ [ ] (Round to three decimal places as needed.)

6. Solve.

   $9^{-x+3} = 82$

   x ≈ [ ]
   (Simplify your answer. Type an integer or decimal rounded to four decimal places as needed.)

7. A minute ad during a campaign in 1973 cost $200,000. The price tag for a 30-second ad slot during a 2006 campaign was $3 million. The cost of a 30-second slot of advertising for a campaign can be modeled by $y = 0.0000967\left(1.103^x\right)$, where x is the number of years after 1900 and y is the cost in millions.

   Complete (a) and (b) below.

   a. According to the model, what was the cost of a 30-second campaign ad in 2003?

   The cost of a 30-second campaign ad in 2003 was $[ ].
   (Simplify your answer. Round to the nearest dollar as needed.)

   b. The cost of a 30-second campaign ad in 2015 was $5.5 million. Does the model underestimate or overestimate the cost of these ads?

   The model (1) ________ the cost for a 30-second campaign ad. The model estimates the cost for 2015 at $[ ].
   (Simplify your answer. Round to the nearest dollar as needed.)

   (1) ○ overestimates
   ○ underestimates